ELSEVIER

Journal of Geochemical Exploration 69–70 (2000) 369–372

JOURNAL OF GEOCHEMICAL EXPLORATION

www.elsevier.nl/locate/jgeoexp

Paleomagnetic dating of fluid-flow events in dolomitized Caledonian basement rocks, central Scotland

R.D. Elmore[a,*], J. Parnell[b], M. Engel[a], S. Woods[a], M. Abraham[a], M. Davidson[a]

[a]*School of Geology and Geophysics, University of Oklahoma, Norman, OK 73119, USA*
[b]*Department of Geology and Petroleum Geology, University of Aberdeen, King's College, Aberdeen AB24 3UE Scotland, UK*

Abstract

Paleomagnetic results suggest that there was a fluid-flow event which caused dolomitization and hematite authigenesis in serpentinite within the Highland Border Group along the Highland Boundary Fault in central Scotland in the late Carboniferous/early Permian. Geochemical data indicate that the fluid was hydrothermal in origin, and in addition to causing hematite authigenesis, the fluid also thermally remagnetized pre-existing magnetite in the serpentinite. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: remagnetization; fluids; fault; paleomagnetism

1. Introduction

In central Scotland, there are numerous exposures of red dolomite in the Caledonian (pre-Devonian) basement. Some are within the Dalradian (neoProterozoic) metasedimentary basement whereas others are along the Highland Boundary Fault (HBF) in the Lower Paleozoic Highland Border Group (HBG). This study uses paleomagnetic analysis to date the fluid migration event responsible for the dolomitization in the HBG and to assess whether the dolomite represents a single episode of fluid-flow and alteration of the basement, or if there were multiple events that are juxtaposed coincidentally.

2. Geological setting

The trace of the HBF can be mapped with confidence for a distance of 230 km from Stonehaven to Arran in central Scotland (Fig. 1). The predominant rocks to the north of the fault are Dalradian metasediments whereas the rocks to the south are continental Devonian (Old Red Sandstone, ORS) sediments. At many places there is a narrow outcrop of the HBG, immediately south of the fault plane. These rocks represent an oceanic or backarc assemblage that has been excised almost completely. It includes several exposures of serpentinite, representing sea-floor basic rocks altered by meteoric fluids (Ikin and Harmon, 1983). In some outcrops the serpentinite is massive. In others it is brecciated, either due to tectonic activity or synsedimentary processes.

Outcrops of the serpentinite exhibit extensive dolomitization and/or silicification. The alteration occurs partly as a massive replacement and partly as thin (from 1 to ~10 cm) veins. Silica and dolomite veins show crosscutting relationships, implying that both types are related to a single, but possibly prolonged, episode of alteration. Massively replaced serpentinite includes brecciated fabrics (Henderson and Fortey,

* Corresponding author. Tel.: + 1-405-325-3253; fax: + 1-405-325-3140.
E-mail address: delmore@ou.edu (R.D. Elmore).

0375-6742/00/$ - see front matter © 2000 Elsevier Science B.V. All rights reserved.
PII: S0375-6742(00)00095-9

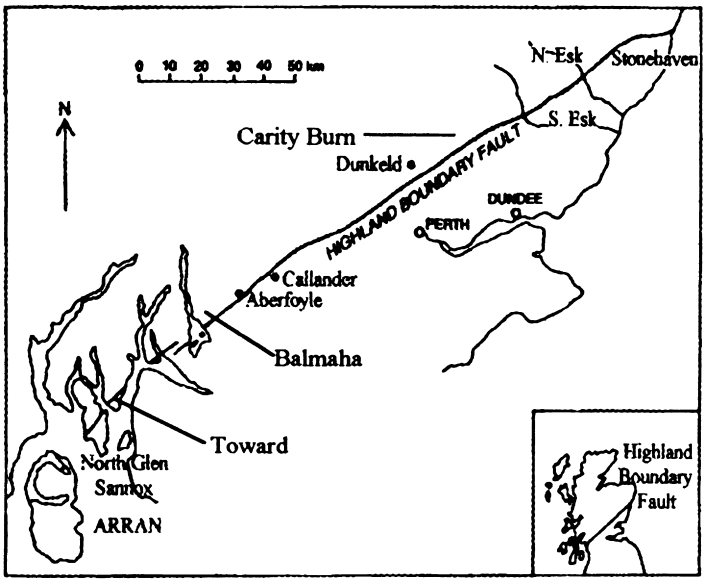


Fig. 1. Location of the three sampling sites along the HBF. Inset map shows the location of the fault in central Scotland.

1982), indicating that replacement was a separate event and post-dates brecciation. There are locations, however, where the altered serpentinite has been faulted and exhibits cataclasis (Allan, 1928; Henderson and Fortey, 1982).

Evidence for the age of the dolomitic and silicic alteration is equivocal. Siliciclastic sediment in the HBF contains clasts with carbonate and quartz, which has led some authors to suggest that the alteration occurred before deposition of these sediments (Henderson and Fortey, 1982). Henderson and Fortey (1982) relate the dolomitization to alteration of talc in the serpentinite by CO_2-rich hydrothermal fluids, as found in other serpentinites (e.g. Groves et al., 1974). Clasts of red dolomitic serpentinite have been found also in Lower ORS conglomerates (Allan, 1928), implying pre-ORS dolomitization. However, alteration extends into Lower ORS sediments (Jehu and Campbell, 1917), and petrographic studies indicate the dolomite has replaced Lower ORS lavas. Thus, there is conflicting evidence for whether the alteration occurred before or after Caledonian orogenic activity.

The hematite, which accounts for the red staining in the serpentinite, appears to be intergrown with the dolomite and apparently formed at the same time. The hematite did not form during dedolomitization (e.g. Elmore et al., 1985).

3. Sampling and processing

Cores for paleomagnetic and rock magnetic studies were collected with a portable drill and oriented with an inclinometer and Brunton compass from the red serpentinite at three localities (Toward, Balmaha, Carity Burn; Fig. 1). Dark green serpentinite was also sampled at two localities along the HBF but only unoriented cores could be retrieved because the rock is so soft. Standard paleomagnetic samples were subjected to alternating field (AF) and thermal demagnetization (e.g. Banerjee et al., 1997) to elicit directional information using Fisher (1953) statistics. Rock magnetic tests were carried out to identify and characterize the magnetic mineralogy (Banerjee et al., 1997). Petrographic analysis of the samples was accomplished using polished thin sections under transmitted and reflected light.

4. Results and interpretations

Thermal demagnetization of most red, dolomitized serpentinite specimens from the three locations removes a characteristic magnetization (ChRM) with southerly declinations and shallow negative inclinations. Removal of a significant fraction of the

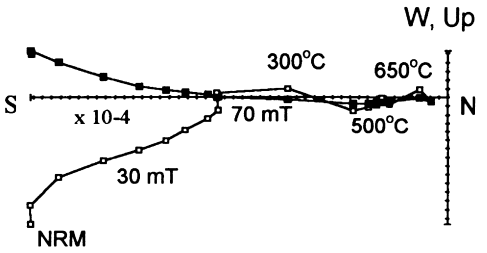


Fig. 2. Orthogonal projection of the demagnetization of a representative specimen revealing a southerly and down component removed by AF treatment and a southerly and shallow component by thermal treatment (open symbols, vertical component; solid symbols, horizontal component).

ChRM between 580 and 680°C suggests that the ChRM resides in hematite. Alternating field demagnetization does not remove significant magnetization in most specimens although in some specimens a component with southerly declinations and shallow inclinations that is similar to the ChRM is removed (Fig. 2). The AF treatment does not completely remove the magnetization and subsequent thermal treatment removes the ChRM (Fig. 2). In contrast to these results from the red serpentinite, the magnetization in the undolomitized green serpentinite is completely removed by AF treatment but since the cores were unoriented, no directional information could be determined.

The acquisition curves of an isothermal remanent magnetization (IRM) for most of the red serpentinite specimens indicates that a high coercivity phase dominates. Decay of the IRM to 680°C suggests that

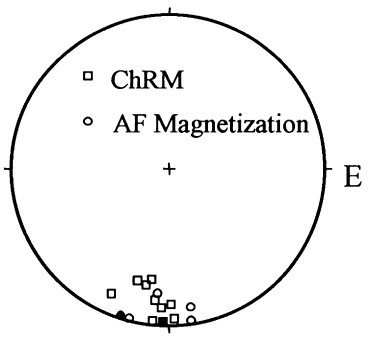


Fig. 3. Equal area plot of site mean directions for both components from the three locations. Average number of specimens per site is 5 (solid symbols, positive inclinations; open symbols, negative inclinations).

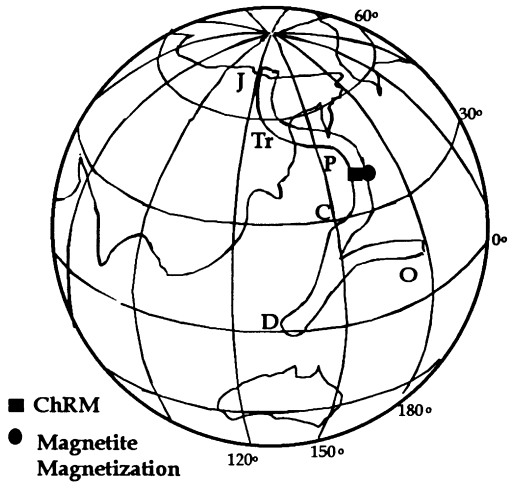


Fig. 4. Apparent polar wander path for Europe with the pole positions for the ChRM and the magnetite magnetization. Both poles are similar and indicate that the magnetizations were acquired in the late Carboniferous/early Permian.

the mineral is hematite. Some IRM acquisition curves of red serpentinite, however, indicate the presence of both low and high coercivity phases. Thermal decay results from these samples are consistent with the presence of both magnetite and hematite. The IRM acquisition curves for the green, relatively unaltered serpentinite suggest that a low coercivity phase dominates the magnetization. Thermal decay by 580°C suggests that the mineral is magnetite.

The results from the red serpentinite indicate the presence of two similar southerly and shallow magnetizations in the red serpentinite (Fig. 3). The mean direction for the ChRM which resides in hematite has a declination (D) of 187.7° and an inclination (I) of −15.9° (cone of 95% confidence $[\alpha_{95}] = 7.5°$; k [grouping] = 34, $n = 11$). The component removed by AF demagnetization which resides in magnetite has a $D = 181.9°$ and $I = -9.4°$ ($\alpha_{95} = 16.9°$; $k = 17.1$, $n = 5$). The pole positions for both magnetizations are similar and suggest they were acquired in the late Carboniferous/early Permian (Fig. 4).

The maximum demagnetization temperature of 680°C for the ChRM is too high for this component to have been acquired by a thermoviscous remagnetization mechanism, and it is therefore interpreted to be a chemical remanent magnetization (CRM) caused by hematite authigenesis. The abundant authigenic

hematite in the altered serpentinite is consistent with this interpretation. This CRM is interpreted to have been caused by the dolomitizing fluids, which moved along the HBF.

The origin of the component removed by AF demagnetization is more problematic. The maximum unblocking temperature for this component cannot be determined because it is always found in specimens with the hematite component. As a result, it is difficult to evaluate the possibility of a heating-related remagnetization mechanism. Although this magnetization could be a CRM, the relatively high temperatures (200°C) recorded in the rocks based on fluid inclusion studies (Parnell et al., 2000a) suggest that it could be a remagnetization caused by heating and cooling (i.e. thermoviscous) of pre-existing magnetite in the serpentinite. In any case, because the magnetization is localized in the HBF within the fault zone, it is probably related to the fluids.

5. Discussion

The paleomagnetic results suggest that there was a fluid-flow event that caused dolomitization and hematite authigenesis along the HBF in the late Carboniferous/early Permian age. The fact that the fluid-related magnetizations are found at three widely separated locations (Fig. 1) along the HBF suggest that the fluid event was widely distributed. Although meteoric fluids may have been involved, the presence of relatively high temperature fluid inclusions in the altered rocks as well as evidence for formation of high temperature clays (Parnell et al., 2000a) suggest that the fluids were hydrothermal in origin. These fluids caused precipitation of hematite and acquisition of the associated CRM. In addition, the hot fluids probably reset a pre-existing magnetization in the serpentinite. These fluids could be related to the intrusion of late Carboniferous dikes in central Scotland (Parnell et al., 2000a) and may be related also to a hypothesized regional scale-flow event that caused remagnetization in Scotland (Torsvik et al., 1989).

The dolomite along the HBF may be related to dolomite in the Dalradian schist in west-central Scotland, which occurs as linear vein-fillings that contain brecciated country rock. Paleomagnetic analysis indicates that hematite authigenesis occurred in the Permo-Triassic, probably due to oxidative weathering (Parnell et al., 2000b). Petrographic evidence, however, indicates that the dolomitization pre-dated the hematite authigenesis. Moderate temperatures (80–115°C) from fluid inclusion studies and stable isotopic data suggest that the dolomitizing fluids were basinal brines (Parnell et al., 2000b). The dolomite veins probably formed during the late Carboniferous extension (Parnell et al., 2000b), and may be related to the dolomite which occurs along the HBF. Petrographic and geochemical studies are underway to assess this possibility.

In summary, the results from the fault zone rocks in the HBF indicate that there was at least one dolomitizing and hematite-precipitating fluid alteration event in central Scotland in the late Carboniferous. This fluid event was widespread and altered rocks along much of the HBF.

References

Allan, D.A., 1928. The Geology of the Highland border from Tayside to Noranside. Trans. R. Soc. Edinb. 56, 57–87.

Banerjee, S., Engel, M., Elmore, R.D., 1997. Chemical remagnetization and Burial Diagenesis of organic matter: Testing the hypothesis in the Pennsylvanian Belden Formation, Colorado. J. Geophys. Res. 102, 24825–24842.

Elmore, R.D., Dunn, W., Peck, 1985. Absolute dating of a diagenetic event using paleomagnetic analysis. Geology 15, 558–561.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London Ser. A 217, 295–305.

Groves, D.I., Hudson, D.R., Hack, T.B.C., 1974. Modification of iron–nickel sulphides during serpentinization and talc-carbonate alteration at Black Swan, Western Australia. Econ. Geol. 69, 1265–1281.

Henderson, W.G., Fortey, N.J., 1982. Highland Border rocks at Loch Lomond and Aberfoyle. Scott. J. Geol. 18, 227–245.

Ikin, N.P., Harmon, R.S., 1983. A stable isotope study of serpentinization and metamorphism in the Highland Border Suite, Scotland, UK. Geochim. Cosmochim. Acta 47, 153–167.

Jehu, T.J., Campbell, R., 1917. The Highland Border rocks of the Aberfoyle district. Trans. R. Soc. Edinb. 52, 175–212.

Parnell, J., Baron, M., Boyce, A., 2000. Regional kaolinization of the Highland Boundary Fault zone and the Caledonian unconformity, Scotland and Northern Ireland. J. Geol. Soc. London (in press).

Parnell, J., Baron, M., Davidson, M., Elmore, D., Engel, M., 2000. Dolomitic breccia veins as evidence for extension and fluid flow in the Dalradian of Argyll. Geol. Mag. (in press).

Torsvik, T.H., Lyse, O., Atteras, G., Blunck, B.J., 1989. Paleozoic paleomagnetic results from Scotland and their bearing on the British apparent polar wander path. Phys. Earth Planet. Int. 55, 93–105.